

DESAMO: A Device for Elder-Friendly Smart Homes Powered by Embedded LLM with Audio Modality

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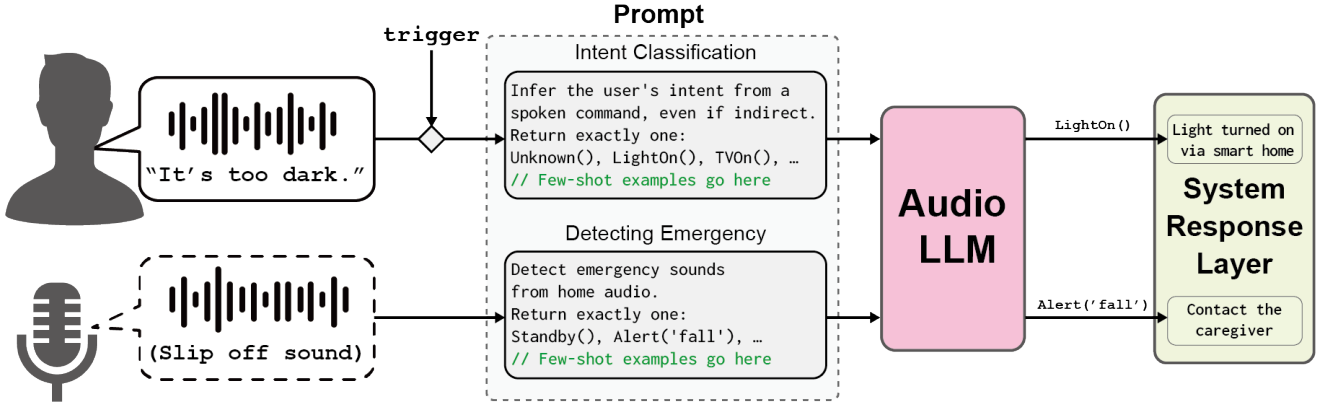


Figure 1: Pipeline Overview of DESAMO

Abstract

We present DESAMO, an on-device smart home system for elder-friendly use powered by Audio LLM, that supports natural and private interactions. While conventional voice assistants rely on ASR-based pipelines or ASR-LLM cascades, often struggling with the unclear speech common among elderly users and unable to handle non-speech audio, DESAMO leverages an Audio LLM to process raw audio input directly, enabling a robust understanding of user intent and critical events, such as falls or calls for help.

CCS Concepts

• Human-centered computing → Sound-based input / output.

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*Both authors contributed equally to this research. Youngwon Choi led the on-device execution of Audio LLM architecture and designed the prompting strategies. Donghyuk Jung developed the interface components and contributed to overall system integration.

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1 Introduction

With the rise of aging populations and single-person households, there has been a steady increase in demand for smart home technologies that support personalized and contactless interaction [4]. Voice assistant interfaces [8, 19], in particular, have become an intuitive means of controlling devices without manual input. However, conventional voice assistant systems typically rely on automatic speech recognition (ASR) based transcription followed by shallow intent parsers, making them poorly suited to interpret indirect expressions and ambiguous language [10]. To address these limitations, recent systems have introduced ASR-LLM pipelines [2], in which transcribed speech is passed to large language models. Nevertheless, these systems remain vulnerable to transcription errors [5], particularly when processing the unclear articulation of elderly users [16], and such systems are not capable of processing non-speech audio inputs. More recently, large language models have been extended to the audio modality (Audio LLM), such as AudioChatLLaMA [6] and Qwen-Audio [3], enabling end-to-end inference directly from both speech and non-speech audio inputs. By jointly reasoning over acoustic and semantic information, these models allow a more robust and flexible audio understanding, eliminating the need for intermediate text representations.

In this work, we propose DESAMO, an audio-interactive smart home system for older adults based on the Audio LLM that runs entirely on-device. DESAMO is capable of understanding not only direct and indirect spoken instructions, but also environmental sounds such as falls and screams, allowing for both voice intent classification and emergency event detection within a single unified model. Figure 1 presents the overall architecture that supports both

Table 1: Performance of Cascaded and Proposed Approaches

Model	WER	Accuracy	Model Size
Whisper medium + LLM	4.55%	96.33%	3.64GB
Whisper large-v3 + LLM	3.89%	97.33%	5.20GB
Proposed	N/A	98%	3.45GB

tasks using a shared Audio LLM. By executing all computations locally on edge hardware, this design inherently enhances user privacy, which is widely recognized as a critical factor in the adoption of smart homes [14]. To the best of our knowledge, this represents one of the first academic demonstrations of a fully on-device Audio LLM implemented as a functional prototype.

2 DESAMO Prototyping

2.1 On-device DESAMO Execution

DESAMO is built upon the Qwen2.5-Omni 3B [18], a recent multimodal language model capable of processing both speech and ambient audio. For audio understanding, this model uses a Whisper large-v3 based audio encoder [12] to transform the raw wave file into semantic embeddings, which are then fed into the language model. The entire system runs locally on NVIDIA Jetson Orin Nano, enabling full inference on-device without any cloud dependency. We use a quantized model with 16-bit audio encoders and a 4-bit language model, packaged in the compact GGUF format.

2.2 Voice Intent Classification

Recent advancements in function calling have enabled language models to generate structured function representations from natural language queries. Prior works [1, 11, 13] demonstrate that even 2B to 7B scale models can effectively retrieve and generate functions based on user queries. In our use case, intent classification can be interpreted as a form of function calling, where natural language inputs like “call my daughter” are mapped to structured commands such as *Call('daughter')*. Inspired by this paradigm, DESAMO extends function calling to the audio domain, allowing users to issue commands using natural speech rather than text-based input.

Upon detecting a trigger phrase, DESAMO records a short audio segment that may contain either a direct command like “turn on the air conditioner” or an indirect expression such as “It’s getting hot.” The speech is processed into a semantic embedding and passed into the intent classification pipeline, along with a prompt that guides the model to generate structured control outputs such as *ACOn()*. The system response layer interprets this output to activate the corresponding device, accompanied by a brief voice confirmation.

2.3 Detecting Emergencies

To ensure safety in environments where aging is an issue, DESAMO uses passive audio monitoring to detect emergencies, such as falls or distress, drawing inspiration from recent advances in prompt-based sound understanding [7, 15] with Audio LLM.

The system continuously monitors ambient audio by capturing short sound segments at regular intervals. At each interval, the system feeds an audio segment and a detection prompt into the Audio LLM to identify potential emergencies, such as falls or verbal cries

>> Audio LLM	
Target intent:	DecreaseHeat()
Predicted intent:	DecreaseHeat() ✓
>> Whisper medium + LLM	
Target transcription:	Decrease the heating
ASR result:	to increase the heating
Target intent:	DecreaseHeat()
Predicted intent:	IncreaseHeat() ✗
>> Audio LLM	
Target intent:	LightOn()
Predicted intent:	LightOn() ✓
>> Whisper large-v3 + LLM	
Target transcription:	Lights on
ASR result:	Bye, Thon.
Target intent:	LightOn()
Predicted intent:	Unknown() ✗

Figure 2: Audio LLM directly predicts intent without suffering from ASR error propagation.

for help, from elderly users. The model outputs structured event labels, such as *Alert('fall')* or *Alert('help')*, which are interpreted by the system response layer to trigger responses, including sending an alert or sounding an alarm, and notifying caregivers.

3 Pilot Evaluation

To evaluate DESAMO in a realistic use case, we built a pilot benchmark using 300 samples curated from the Fluent Speech Commands dataset [9], filtering for speakers aged 65 years or older, and selecting to match the voice intent classification scenario of DESAMO. We compared DESAMO with two cascaded baselines: one combining Whisper large-v3 with Qwen2.5-Omni in text-only mode, and another using Whisper medium with the same LLM. Whisper medium was selected because its parameter size is comparable to that of the Whisper large-v3 encoder, and both models were quantized to 16-bit to ensure a fair comparison with the Qwen2.5-Omni audio encoder. As shown in Table 1, DESAMO achieved the highest intent classification accuracy, outperforming both baselines while using the smallest total model size involved in inference. We observed that ASR errors in cascaded systems often led to intent misclassification, as illustrated by the examples in Fig. 2. In contrast, the Audio LLM in DESAMO correctly handled these cases by avoiding reliance on intermediate transcripts. Since the Fluent Speech Commands dataset is relatively clean in terms of pronunciation and recording conditions, we expect this performance gap to widen further in more realistic and noisy environments.

As a side note, we found that the system also handled non-English voice commands without any modification of the prompt, aligning with the cross-lingual generalization patterns observed in prior work [17].

4 Conclusion and Future Work

We present DESAMO, an embedded smart home system that uses Audio LLMs to support elderly users through natural voice-based control and passive monitoring of critical ambient events, offering robust interaction entirely on edge hardware without compromising privacy. Beyond elderly care, this approach opens up possibilities for broader applications of edge-based Audio LLMs in contexts

where privacy or network access are constrained. For future work, we plan to extend the system to multimodal settings by utilizing visual input for richer context understanding and to optimize inference latency — currently around 5.3 seconds in headless mode — for more responsive on-device interaction.

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